ONE-PAGE BRIEF

Post-quantum cryptography review: fabric-lib-go @ f8b4f4a833e4

Client: Hyperledger Fabric (LF Decentralized Trust) — public repository (passive review) · Source: github.com/hyperledger/fabric-lib-go, commit `f8b4f4a833e4` · Reviewed: 2026-11-06 10:15 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY
Level 0 of 4
no post-quantum cryptography

FINDINGS
0 Critical **1** High **3** Medium
0 Low **7** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)
Key establishment: 10 + 5 − 8 = 7 → **too late**
Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

WHAT MATTERS MOST

High QSI-2026-17514	**google.golang.org/grpc v1.83.1 is affected by GHSA-2v4p-qf9q-27wj (CVE-2026-84445, GO-2026-6443)** What to do: Upgrade to a fixed version (1.82.2, 1.83.2, 1.84.0-dev.0.20260825144003-d5a41119e0e3, 1.85.0-dev.0.20260825072537-93e31b48545e).
Medium QSI-2026-17511	**Quantum-vulnerable ECDSA: role not determined (10 file(s))** preliminary What to do: Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form...
Medium QSI-2026-17512	**Quantum-vulnerable Ed25519: role not determined (6 file(s))** preliminary What to do: Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form...
Medium QSI-2026-17513	**Quantum-vulnerable RSA: role not determined (6 file(s))** preliminary What to do: Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form...

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium role not determined	ECDSA, Ed25519, RSA	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)...

KNOWN VULNERABLE COMPONENTS

Advisories for the versions in use: 8; unlikely to apply to this build: 7 (a Cargo feature that is off, affected functions not named in the code, or a platform that is not built); to review: 1.

Static review of source code and manifests; files analysed: 67. No requests were sent to the owner's systems.

Full report, content SHA-256: `967ff3502b91c3f489ae4dc96bfd80d7f227c73a45038ab60ffb4c654876bd06`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.